



A Real-Time Wildfire Resilience Digital Twin

Group: N-Dimensional
Group Number: 3

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Background & Motivation

The Threat

Wildfires increasingly threaten infrastructure, ecosystems, and human safety, particularly in urban-dense regions like California.

The Gap

Current monitoring systems often lack real-time data integration and interactive decision-making capabilities.

The Solution

A "Digital Twin" that mirrors real-world environments in real time to provide updated risk representations.





FOCUS

Project Overview & Big Data Relevance

System Goal

Support rapid situational awareness through streaming data ingestion and geospatial analytics.

Volume

Processing over 11.5 million building polygons for the entire state of California.

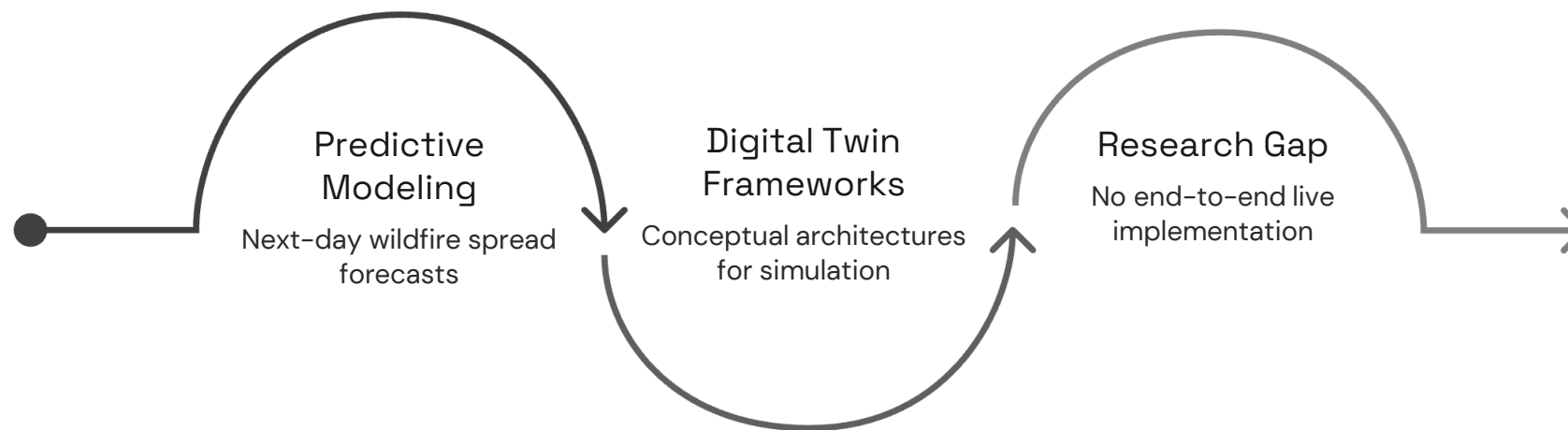
Velocity

Ingesting high-velocity fire events via Apache Kafka streaming pipeline.

Variety

Fusing dynamic sensor feeds (temperature, wind) with static infrastructure data.

Related Work



The existing body of work is divided between predictive models and theoretical frameworks, leaving a gap for concrete, streaming-backed implementations.

Current Landscape

- Predictive modeling approaches (e.g., Next Day Wildfire Spread)
- Conceptual digital twin frameworks without live implementation
- No end-to-end system coupling visualization to a streaming backend

Our Contribution

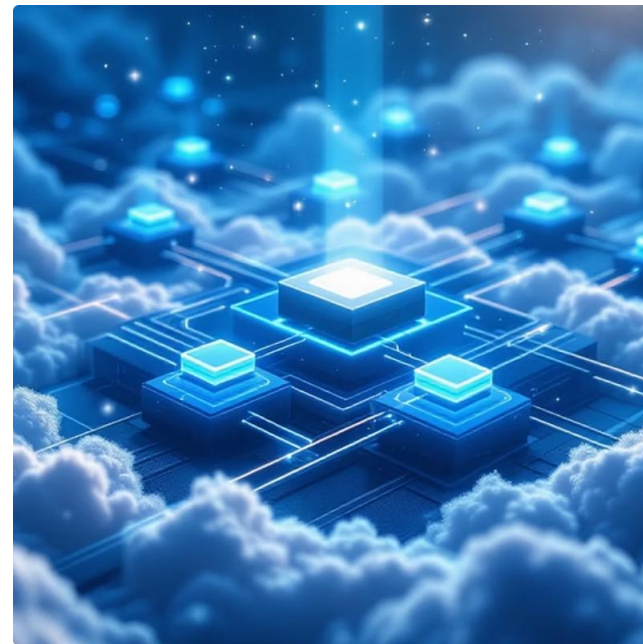
A concrete, end-to-end implementation connecting live event ingestion to an infrastructure-aware dashboard with multi-resolution context.

System Architecture



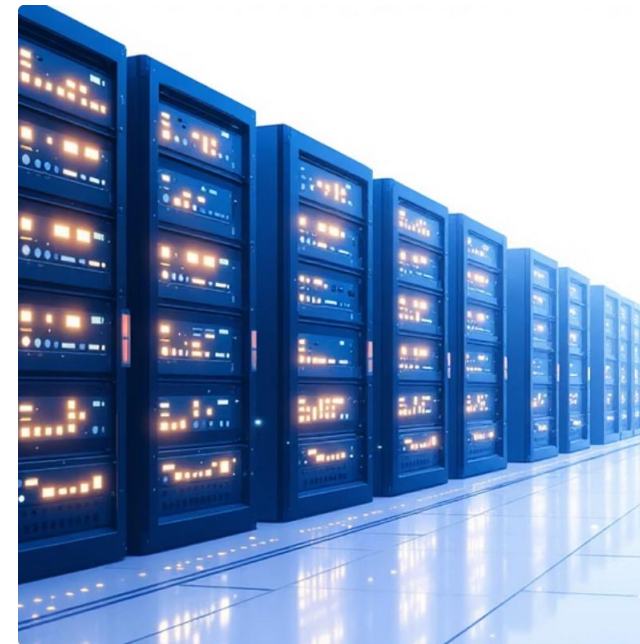
Ingestion

Kafka topic `fire_events` captures high-velocity fire event streams in real time.



Processing

Apache Spark & Sedona perform spatial cross-referencing against building and POI data.



Storage

DuckDB for live alerts; GeoParquet for static building context across California.



Visualization

A real-time dashboard renders fire risk zones, building alerts, and infrastructure context.



MAIN WORK

Data Engineering & Enrichment



11.5M Building Footprints

Converted Microsoft building polygons to Sedona-compatible GeoParquet format.



Semantic Join

ST_Contains used to join building polygons with OpenStreetMap POIs.



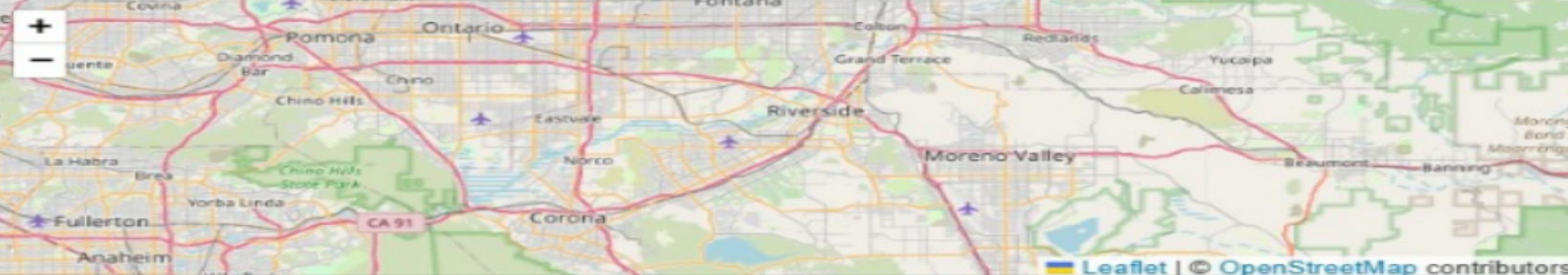
~80,000 Essential Buildings

Hospitals, schools, and emergency services identified for prioritized risk analysis.

Key Outcome

Infrastructure-Aware Risk

Essential buildings are pre-classified, enabling the system to immediately surface the most critical at-risk assets during any fire event.



MAIN WORK

Real-Time Risk Modeling



Geospatial Reasoning

Incoming fire events are transformed into geometric risk regions projected onto the map.



Directional Model

Evolved from simple circular buffers to cone-shaped regions influenced by wind speed and direction.



Infrastructure Intersection

System identifies infrastructure within the predicted spread zone and generates prioritized alert records.



EVALUATION

Evaluation Methodology & Scalability

<5s

City-Scale Load

Buildings + alerts rendered in under 5 seconds for a full city-scale view.

11.5M

Building Polygons

Statewide California dataset processed with H3 indexing for multi-resolution access.

~80K

Essential Buildings

Critical infrastructure pre-identified for fast prioritized alerting.

H3 Indexing

Statewide data aggregated into coarse cells; finer details load only when zooming in, reducing render overhead dramatically.

Partition Pruning

Restricted GeoParquet reads to specific spatial regions, avoiding full dataset scans and speeding up queries.

Simulation & Evaluation

Scenario Testing

Validated via interactive simulations where users inject fire events with custom weather data parameters.

Results

System successfully generated intuitive risk regions that correctly highlighted at-risk infrastructure in the dashboard across all tested scenarios.

Responsiveness

Decoupling DuckDB from Kafka ensures the UI remains stable and performant during high-throughput ingestion bursts.



Conclusion & Future Directions

1

Live Data

Transition from simulated events to live NASA FIRMS fire data feeds

2

Terrain Dynamics

Incorporate terrain elevation and fuel load data for higher-fidelity spread modeling

3

Decision Support

Expand "what-if" analysis tools for emergency planners and decision-makers

Key Takeaways

Built a scalable, multi-resolution digital twin using a modern big data stack: Kafka, Spark, Sedona, DuckDB

Enables "what-if" analysis for decision-makers to explore potential fire impacts under varying conditions

End-to-end pipeline from live event ingestion to infrastructure-aware geospatial dashboard

Thank You

Questions and Discussion